



Operations Research

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Solution to the Training Exercises, Sheet 7

Training Exercises

Exercise T7.1. *Cutting*

Problem

A carpentry shop receives an order from a customer to produce 30 boards of length $5m$ (type A), 10 boards of length $10m$ (type B) and 13 boards of length $12m$ (type C). She has $30m$ length boards available for cutting. She incurs a cost of 1 Euro per board used. Due to technical limitations, additional costs are incurred in the following cases:

- If the cutting pattern contains more than 2 boards of length $5m$.
- If the cutting pattern contains more than 1 board of length $10m$.

For each case that applies, there is an additional cost of 1 Euro. The carpentry would like to solve the problem by means of column generation.

- a) For the CG algorithm, an optimal cutting pattern must be generated in each iteration with respect to a given utility vector $u = (u_A, u_B, u_C) \in \mathbb{R}^3$ and the costs described above. Let u_T denote the utility of a board of type $T \in \{A, B, C\}$. Formulate an IP that solves the problem "Find an optimal cutting pattern with respect to u " and implement it in Gurobi.
- b) Denote by x_1, x_2, x_3 the initial basis, where x_1 is the pattern with six $5m$ boards, x_2 is that with three $10m$ boards, and x_3 is that with two $12m$ boards. Indicate which basis you get after 2 iterations of the CG algorithm.

Solution

- a) A cutting pattern is given by the variables $y_T \in \mathbb{N}_0$, which indicate the number of boards of type $T \in \{A, B, C\}$ in the cutting pattern. Let A be the indicator for $y_A \geq 3$

and B be the indicator for $y_B \geq 2$. Then we can calculate a solution by solving

$$\begin{aligned} \max \quad & \sum_T u_T y_T - A - B - 1 \\ \text{s.t.} \quad & 4A \geq y_A - 2 \\ & 2B \geq y_B - 1 \\ & 5y_A + 10y_B + 12y_C \leq 30 \\ & y_T \in \mathbb{N}_0 \\ & A, B \in \{0, 1\} \end{aligned}$$

- b) 1. *Iteration* The basic variables are x_1, x_2 and x_3 as specified. According to the specification, $c_B = (2, 2, 1)^T$, because pattern 1 cuts out more than two 5m boards and pattern 2 cuts out more than one 10m board. The constraint matrix for this basis is given by

$$B = \begin{pmatrix} 6 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

and the right side is $b = (30, 10, 13)^T$. Further we have

$$B^{-1} = \begin{pmatrix} 1/6 & 0 & 0 \\ 0 & 1/3 & 0 \\ 0 & 0 & 1/2 \end{pmatrix}$$

and, therefore,

$$c_B^T B^{-1} = (1/3, 2/3, 1/2).$$

Thus, to compute a new basic variable, we need to solve the above IP with utility vector $u = (1/3, 2/3, 1/2)$. An optimal solution is given by the pattern $y = (1, 1, 1)^T$. Let x_4 denote the number of boards produced according to this cutting pattern. It holds that $b' = B^{-1}b = (5, 10/3, 13/2)^T$ and $y' = B^{-1}y = (1/6, 1/3, 1/2)^T$. We get $b'/y' = (30, 10, 13)$ and, consequently, basis variable x_2 is replaced by x_4 .

2. *Iteration* The constraint matrix for the new basis $\{x_1, x_4, x_3\}$ is given by

$$\begin{pmatrix} 6 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 2 \end{pmatrix}.$$

The new cost vector is $c_B^T = (2, 1, 1)$. It holds that

$$B^{-1} = \begin{pmatrix} 1/6 & -1/6 & 0 \\ 0 & 1 & 0 \\ 0 & -1/2 & 1/2 \end{pmatrix}$$

and, therefore,

$$c_B^T B^{-1} = (1/3, 1/6, 1/2).$$

With respect to this utility vector, we solve the IP and obtain $y = (1, 0, 2)^T$ as the new cutting pattern. Let x_5 be the number of boards cut according to this pattern. It follows that $b' = B^{-1}b = (10/3, 10, 3/2)^T$ and $y' = B^{-1}y = (1/6, 0, 1)^T$. Thus $b'/y' = (20, \infty, 3/2)$ and the outgoing basic variable is consequently x_3 . Thus, the basis after two iterations is given by $\{x_1, x_4, x_5\}$.

Exercise T7.2. Hiking trip

Problem

Joe Hännies is organizing a hike with his friends. He wants to take bread with different spreads with him. There are a total of $m \in \mathbb{N}$ different spreads and he wants to take at least b_i loaves with spread i with him. He has identical boxes at his disposal for transportation. As Joe Hännies is a gourmet, he pays attention to how well spreads harmonize aromatically. For the spread i , let $H_i \subseteq \{1, \dots, m\}$ be the quantity of spreads that harmonize with i . Then a filling pattern of a box is only *valid*, if the following condition is met for all spreads i : The number of loaves with i harmonizing spreads is at least as large as the number of loaves with spread i . Furthermore each box can hold a maximum of G loaves and no box may remain empty.

For example, let's assume $H_1 = \{5, 8\}$ and fill a box with two loaves of bread with spread 1. Then, according to the above condition, at least two more loaves with spread 5 or 8 would have to be added to the box.

Joe Hännies wants to take as few boxes as possible with him and wants to solve this problem using column generation. To do this, he first sets up the following IP:

$$\begin{aligned} \min \quad & \sum_{i=1}^n c_i x_i \\ \text{s.t.} \quad & Ax = b \\ & x \in \mathbb{N}_0^n. \end{aligned}$$

The matrix A has m rows and n columns, where each column describes an admissible filling pattern of a box. Furthermore, $c = (1, \dots, 1)^T \in \mathbb{R}^n$. The column $a \in \mathbb{N}_0^m$ therefore describes a box in which there are exactly a_i loaves with spread i . Furthermore, x_i indicates how often the filling pattern in the i -th column is used.

- a) Describe the condition " $a = (a_1, \dots, a_m)^T \in \mathbb{N}_0^m$ is valid" by linear (in-)equalities.
- b) In this subtask we assume that $G = 3$ and $m = 2k$ for $k \in \mathbb{N}$ is arbitrary. Specify the number of valid filling patterns for each of the following two cases:
 - Case 1: $H_i = \{i\}$ for all $i \in \{1, \dots, m\}$
 - Case 2: $H_i = \begin{cases} \{i+1\}, & \text{for } i \text{ odd} \\ \{i-1\}, & \text{for } i \text{ even} \end{cases}$
- c) Explain in which of the two cases the use of Column Generation is particularly suitable?
- d) For this subtask, we assume that $m = 4$, $G = 4$ and $H_1 = \{1\}$, $H_2 = \{2\}$, $H_3 = \{2, 4\}$, $H_4 = \{4\}$. Specify the pricing problem and an optimal solution for the first iteration of the column generation algorithm (the outgoing base variable does not have to be determined). Start with the following base:

$$\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

Solution

- a) At least one loaf in the box:

$$\sum_{i=1}^m a_i \geq 1$$

The total size of the box is not exceeded:

$$\sum_{i=1}^m a_i \leq G$$

The amount of spreads harmonizing with i is at least as large as the number of breads with spread i :

$$\sum_{j \in H_i} a_j - a_i \geq 0 \forall 1 \leq i \leq m$$

- b) Case 1: Each spread only harmonizes with itself, so the constraint due to the quantities H_i is omitted: $\sum_{j \in H_i} a_j - a_i = a_i - a_i \geq 0$. There must be at least one loaf, but there can be a maximum of three loaves in a box. This corresponds to drawing 1, 2 and 3 times from an urn with m balls with putting back. There are thus $m + \binom{m+1}{2} + \binom{m+2}{3}$ valid filling patterns.

Case 2: A filling pattern is valid if and only if there is at least one bread in the box and another bread with the corresponding harmonizing spread. To see this, note that harmonizing spreads always appear in pairs here. So we can only ever put matching pairs in a box. Since $G = 3$, we can only ever put exactly one pair in a box. The number of valid filling patterns is therefore $m/2$.

- c) The use of the CG algorithm is particularly useful if the number of columns significantly exceeds the number of rows of A . The CG algorithm is therefore particularly suitable for the first case.
- d) The base matrix and its inverses are given by :

$$B = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, B^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

The pricing problem is therefore

$$\begin{aligned} & \max c_B^T B^{-1} a - c_a \\ & = \max(1, 1, 0, 1)^T a - c_a \\ & = \max a_1 + a_2 + a_4 - 1 \\ & s.t. \ a_1 + a_2 + a_3 + a_4 \geq 1 \\ & \quad a_1 + a_2 + a_3 + a_4 \leq 4 \\ & \quad a_2 + a_4 - a_3 \geq 0 \\ & \quad a_1, a_2, a_3, a_4 \in \mathbb{N}_0 \end{aligned}$$

The optimal solutions are:

$$a^* \in \left\{ \begin{pmatrix} 4 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ 0 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 3 \\ 0 \\ 0 \end{pmatrix}, \right. \\ \left. \begin{pmatrix} 1 \\ 2 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \\ 3 \end{pmatrix}, \begin{pmatrix} 0 \\ 4 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 3 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 2 \\ 0 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 3 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 4 \end{pmatrix} \right\}.$$

Self-study Exercises

Exercise S7.1. Cutting

Problem

A carpentry shop receives an order from a customer to produce 16 boards of length $4m$, 7 boards of length $7m$, and 13 boards of length $9m$. Fabienne is responsible for the order and has $19m$ length boards available for cutting. She incurs a cost of 1 Euro per board used. Her colleagues tell her to consider the following cutting patterns, where $a_1 = (a_{11}, a_{12}, a_{13})^T = (3, 0, 0)^T$ contains three boards of length $4m$, $a_2 = (a_{21}, a_{22}, a_{23})^T = (0, 2, 0)^T$ contains two boards of length $7m$, and $a_3 = (a_{31}, a_{32}, a_{33})^T = (0, 0, 1)^T$ contains one board of length $9m$. Let furthermore denote with $x_1, x_2, x_3 \in \mathbb{N}_{\geq 0}$ how often each cutting pattern is used. This results in the IP:

$$\begin{aligned} (M) \quad & \min x_1 + x_2 + x_3 \\ & s.t. \quad 3x_1 \geq 16 \\ & \quad \quad 2x_2 \geq 7 \\ & \quad \quad x_3 \geq 13 \\ & \quad \quad x_1, x_2, x_3 \in \mathbb{N}_{\geq 0} \end{aligned}$$

- a) Fabienne thinks the cutting patterns a_1, a_2, a_3 are suboptimal. How and why can she use the column generation algorithm to show her colleagues that she is right?
- b) Specify the pricing problem and an optimal solution for the first iteration of the column generation algorithm (the outgoing base variable does not have to be determined). Start with the base a_1, a_2, a_3 .
- c) Using your results from part b), state the adapted version M' of M with the newly generated column.
Note: You may use placeholder variables if you did not solve part b).

Solution

- a) The CG algorithm searches for the cutting pattern that optimally improves the objective with respect to the shadow prices under the current basis. That is, CG finds the steepest edge (non-basis variable) from the current vertex (basis) in terms of the objective value. If the newly found cutting pattern strictly improves the objective, Fabienne knows that the initial cutting patterns were suboptimal.
- b) The pricing problem is given by

$$\begin{aligned} (P) \quad & \max c_B^T B^{-1} a_4 - c_{a_4} \\ & s.t. \quad 4a_{41} + 7a_{42} + 9a_{43} \leq 19 \\ & \quad \quad a_{41}, a_{42}, a_{43} \in \mathbb{N}_{\geq 0} \end{aligned}$$

where $a_4 = (a_{41}, a_{42}, a_{43})$ denotes the variables describing the novel cutting pattern. As each board costs 1 Euro, we have $c_B = (1, \dots, 1)^T$ and $c_{a_4} = 1$. The basic variables

x_1, x_2, x_3 are as specified above. The constraint matrix for this basis is given by

$$B = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

and the right hand side is $b = (16, 7, 13)^T$. We have

$$B^{-1} = \begin{pmatrix} 1/3 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

and, therefore,

$$c_B^T B^{-1} = (1/3, 1/2, 1).$$

The pricing problem is then

$$\begin{aligned} (P) \quad & \max 1/3a_{41} + 1/2a_{42} + a_{43} - 1 \\ & s.t. \quad 4a_{41} + 7a_{42} + 9a_{43} \leq 19 \\ & \quad \quad a_{41}, a_{42}, a_{43} \in \mathbb{N}_{\geq 0} \end{aligned}$$

with an optimal solution of $a_4^* = (0, 0, 2)^T$.

c) Using the newly computed column from part b) gives us

$$\begin{aligned} (M') \quad & \min x_1 + x_2 + x_3 + x_4 \\ & s.t. \quad 3x_1 \geq 16 \\ & \quad \quad 2x_2 \geq 7 \\ & \quad \quad x_3 + 2x_4 \geq 13 \\ & \quad \quad x_1, x_2, x_3, x_4 \in \mathbb{N}_{\geq 0} \end{aligned}$$

Exercise S7.2. Bin Packing

Problem

We consider the bin packing problem: Given n items $i \in N = \{1, \dots, n\}$ with weights g_i . These items are to be distributed among bins, with items of total weight G fitting into each bin. The number of bins used is to be minimized.

- For a given instance of the bin packing problem, the columns of the matrix $A \in \{0, 1\}^{n \times \alpha}$ encode all α filling patterns of a bin. If a is a column of A , then $a_i = 1$ exactly when there is item i in the corresponding pattern in the bin. Formulate the bin packing problem using A as IP.
- Now let $n = 10$ objects be given, where $g_i = i$. Each bin can further hold a weight of $G = 10$. Perform an iteration of the CG algorithm applied to the LP relaxation of the bin packing problem. Start with the base where each item goes into its own bin.

Solution

- a) For $j \in \{1, \dots, \alpha\}$ denote with $x_j \in \mathbb{Z}_{\geq 0}$, how often the filling pattern in the j 'th column of $A = (a_{ij})$ is used. We get the following IP

$$\begin{aligned} \min & \sum_{j=1}^{\alpha} x_j \\ \text{s.t.} & \sum_{j=1}^{\alpha} a_{ij} x_j \geq 1 \quad \forall i = 1, \dots, n \\ & x_j \in \mathbb{Z}_{\geq 0} \quad \forall j = 1, \dots, \alpha. \end{aligned}$$

(We could also require $x_j \in \{0, 1\}$ here, since there is obviously no point in using the same pattern twice.)

- b) Since we start with each object getting its own bin, $B = I_{10} \in \mathbb{R}^{10 \times 10}$ is the unit matrix. Since $c = (1, \dots, 1)$, we get $c_B^T B^{-1} = (1, \dots, 1) \in \mathbb{R}^{10}$. So, as in the lecture, we need to find a filling pattern a that is optimal with respect to the coefficients $(1, \dots, 1)$:

$$\begin{aligned} \max & \sum_{i=1}^{10} 1 \cdot a_i - 1 \\ \text{s.t.} & \sum_{i=1}^n i a_i \leq G \\ & a_i \in \{0, 1\} \end{aligned}$$

An optimal solution is $a = (1, 1, 1, 1, 0, \dots, 0)$ (since obviously at most 4 items fit in the bin) with value $3 > 0$. To determine the outgoing basis variable, we compute $b' = B^{-1}b = (1, \dots, 1)$ and $N' = B^{-1}a = (1, 1, 1, 1, 0, \dots, 0)$. Since $b'/N' = (1, 1, 1, 1, \infty, \dots, \infty)$, we can choose any of the first 4 variables as the outgoing base variable. Thus, for example, if we choose x_1 , after one iteration we obtain the base matrix B_1 , which emerges from B by swapping the first column by a .